



Department of
Electrical & Electronics Engineering
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Project Report

EE408 Microwave Engineering

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OBJECTIVE

In this study, the dimensions of a 50 Ω microstrip line for 1.4 GHz on an FR-4 substrate were calculated analytically and modelled in CST. Then, a complex load was connected and the mismatch was observed, and a single shunt open-circuit stub was designed and integrated into the model to minimize reflections at 1.4 GHz by matching the complex load impedance to 50 Ω . The results were evaluated using S-parameters, input impedance, and VSWR, and the theoretical and simulation outputs were compared.

BACKGROUND

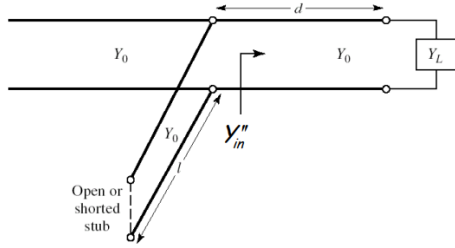


Figure 1: Single-stub tuning circuits

When a transmission line is terminated with a complex load impedance that is different from its characteristic impedance, an impedance mismatch occurs. This causes part of the signal power to be reflected from the load, increases the VSWR, and reduces the power delivered to the system. In this project, the single-stub shunt tuning technique is applied to remove this mismatch. In this method, an open-ended microstrip stub connected in parallel

to the main transmission line is placed at a certain distance from the load to adjust the admittance seen on the line, and the goal is to match the total input impedance to 50 Ω at the target frequency. The single-stub shunt matching method is used to match a complex load $Z_L = R_L + jX_L$ to a transmission line with characteristic impedance Z_0 at a target frequency. The main line is moved a distance d away from the load so that the transformed admittance becomes $Y(d) = G + jB$. At a distance d from the load toward the generator, the impedance becomes: $Y_0 = \frac{1}{Z_0}$.

$$Z(d) = Z_0 \frac{(R_L + jX_L) + jZ_0 \tan(\beta d)}{Z_0 + j(R_L + jX_L) \tan(\beta d)}$$

The admittance at this point is: $Y(d) = \frac{1}{Z(d)} = G + jB$

Here, when the variable transformation $t = \tan(\beta d)$ is applied, the real (G) and imaginary (B) parts of the admittance are calculated in terms of the load impedance parameters (R_L , X_L) using the following formulas:

$$G = \frac{(1 + t^2)}{R_L^2 + (X_L + Z_0 t)^2}, \quad B = \frac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0 [R_L^2 + (X_L + Z_0 t)^2]}$$

Solving for t gives: $t = \frac{X_L \pm \sqrt{\frac{R_L [(Z_0 - R_L)^2 + X_L^2]}{Z_0}}}{R_L - Z_0} \quad \text{for } R_L \neq Z_0$

Based on the obtained t value, the distance of the stub from the load is obtained in terms of wavelength (λ) as follows:

$$\frac{d}{\lambda} = \begin{cases} \frac{1}{2\pi} \tan^{-1}(t), & t \geq 0 \\ \frac{1}{2\pi} (\pi + \tan^{-1}(t)), & t < 0 \end{cases}$$

For the fitting to be complete, the susceptance B_s of the stub must eliminate the imaginary portion of the line $B_s = -B$. For open-circuit stubs, the stub length (l_o) is calculated using the following formula:

$$\frac{\ell_o}{\lambda} = -\frac{1}{2\pi} \tan^{-1} \left(\frac{B}{Y_0} \right)$$

If the length obtained from equation is negative, $\lambda/2$ is added to the result to obtain a physically realizable positive length. Analytical calculations were performed using these formulas and then compared with Smith-Carth.

DESIGN AND TEST PROCEDURES

Analytical calculations were performed using appropriate formulas, in accordance with the specifications requested in the project file. The formulas were calculated as follows, and the theoretical values of the microstrip track were determined.

$$A = \frac{50}{60} \sqrt{\frac{4.3+1}{2}} + \frac{4.3-1}{4.3+1} \left(0.23 + \frac{0.11}{4.3} \right) = 1.51$$

$$\frac{w}{d} = \frac{8e^A}{e^{2A}-2} = \frac{8e^{1.51}}{e^{2 \times 1.51}-2} = 1.95, \text{ so } wd \leq 2w = 1.95 \times 1.53 \text{ mm} = 2.98 \text{ mm}$$

$$\epsilon_e = \frac{4.3+1}{2} + \frac{4.7-1}{2} \times \frac{1}{\sqrt{1+12 \times \frac{1}{1.95}}} = 3.28$$

$$Z_0 = \frac{120\pi}{\sqrt{3.28} * [1.95 + 1.393 + 0.667 * \ln(1.95 + 1.44)]} = 50.13\Omega$$

Z_0 was calculated using formulas and found to be 50.13 ohms. This represents a deviation of 0.26% from the actual impedance value. Based on these results, a microstripline design was created. This design was then optimized for low loss.

A. Analytical Design

Firstly, $Z_L = 42 + j4 \Omega$ and $Z_0 = 50\Omega$ are taken. Finding the $t = \tan(\beta d)$ values:

$$t = \frac{4 \pm \sqrt{\frac{42[(50-42)^2 + 4^2]}{50}}}{42 - 50}$$

$$t = \frac{4 \pm \sqrt{67.2}}{-8} \text{ so } t_1 = \frac{4 + 8.19}{-8} = -1.52 \text{ and } t_2 = \frac{4 - 8.19}{-8} = -0.52$$

Calculation of stub $\frac{d}{\lambda}$ distance: $\frac{d_1}{\lambda} = \frac{\pi - \arctan(-1.52)}{2\pi} \approx 0.342, d_1 = 0.342\lambda$

$$\frac{d_2}{\lambda} = \frac{\arctan(0.52)}{2\pi} \approx 0.076, d_2 = 0.076\lambda$$

Find the susceptance B values:

- For $t_1 = -1.52$: $B_1 = \frac{42^2(-1.52) - (50 - 4(-1.52))(4 + 50(-1.52))}{50[42^2 + (4 + 50(-1.52))^2]} = 0.0039S$
- For $t_2 = 0.52$, since $B_s = -B$, it becomes $B_2 = 0.0039S$. To find the open-loop stub lengths:
 $Y_0 = \frac{1}{50} = 0.02S$

$$\frac{L_{o1}}{\lambda} = -\frac{1}{2\pi} \arctan\left(\frac{0.0039}{0.02}\right) = 0.0306\lambda \quad \frac{L_{o2}}{\lambda} = -\frac{1}{2\pi} \arctan\left(\frac{-0.0039}{0.02}\right) = -0.03\lambda$$

For a negative result, add $+0.5\lambda$, $L_{o2} = (-0.03 + 0.5)\lambda = 0.469\lambda$. Finally, the results are $\lambda = 118.5 \rightarrow d_1 = 40.2\text{mm}$, $L_{o1} = 3.55\text{mm}$ and $d_2 = 9\text{mm}$, $L_{o2} = 55.5\text{mm}$

B. Smith Chart Design

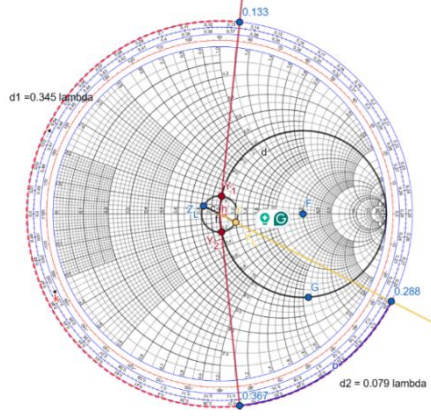


Figure 2: Smith chart solutions for d

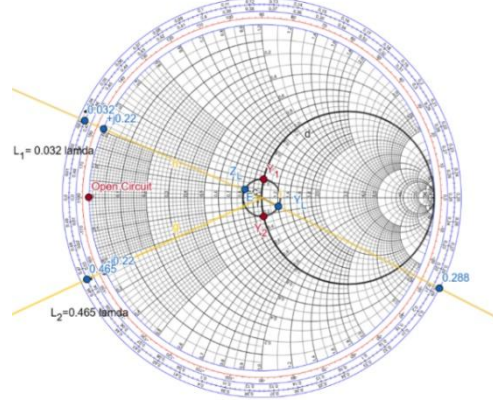


Figure 3: Smith chart solutions for L

Employing smith chart, we successfully attained 2 solutions for placing an open circuit shunt stub. The results of this process are: $d_1 = 0.325\lambda = 38.5\text{mm}$, $L_1 = 0.032\lambda = 3.79\text{mm}$, $d_2 = 0.079\lambda = 9.36\text{mm}$, $L_2 = 0.465\lambda = 55.1\text{mm}$ which are very close to the calculated values.

C. Design Test Procedures

The microstrip line is modelled firstly to match 50Ω impedance.

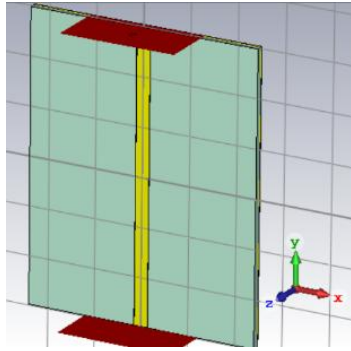


Figure 4: Microstrip Line

Bricks	Min X	Max X	Min Y	Max Y	Min Z	Max Z
GND	$-L_x/2$	$L_x/2$	$-L_y/2$	$L_y/2$	$-h-t$	$-h$
FR4	$-L_x/2$	$L_x/2$	$-L_y/2$	$L_y/2$	$-h$	0
microstrip	$-w/2$	$w/2$	$-L_y/2$	$L_y/2$	0	t

The measurements are finalized as $L_x=40\text{mm}$, $L_y=59.6\text{mm}$, $h=1.53\text{mm}$, $w=2.87\text{mm}$, $t=0.035\text{mm}$ to get the most ideal design.

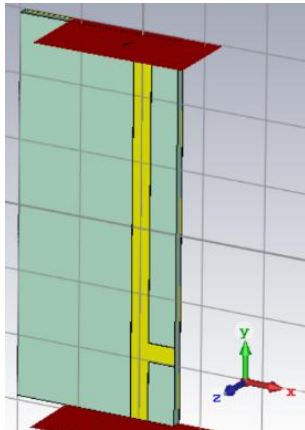


Figure 5: Microstrip w/ stub

Bricks	Min X	Max X	Min Y	Max Y	Min Z	Max Z
GND	$-L_x/2$	$L_x+w/2$	$-L_y/2$	$L_y/2$	$-h-t$	$-h$
FR4	$-L_x/2$	$L_x/2$	$-L_y/2$	$L_y/2$	$-h$	0
microstrip	$-w/2$	$w/2$	$-L_y/2$	$L_y/2$	0	t
stub	$w/2$	$w/2+L$	$L_y/2-d-w/2$	$L_y/2-d+w/2$	0	t

Where $L_x=40\text{mm}$, $L_y=58.6\text{mm}$, $L=4.3$, $d=48.2$ as the idealized version.

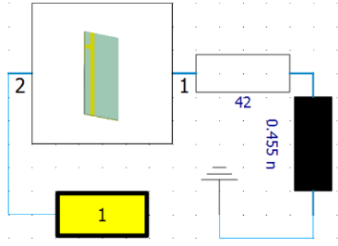


Figure 6: Schematic of the load impedance $Z_L = 42 + j4 \Omega$

The circuit in Figure 6 is constructed to simulate our load (42Ω resistor and 0.455nH inductor) and get the S_{11} parameters as well as line impedance and VSWR in the schematic panel next to 3D simulation panel.

RESULTS AND DISCUSSION

1. Microstrip Line

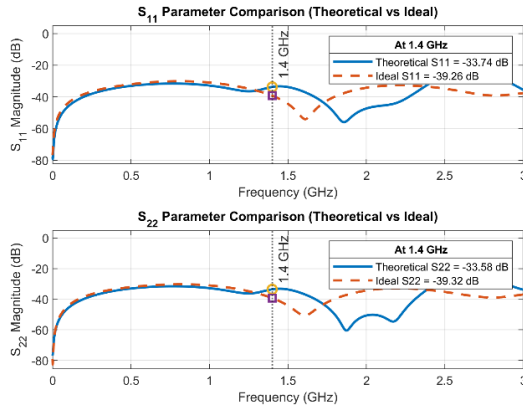


Figure 7: Comparison of S_{11} & S_{22} Magnitude

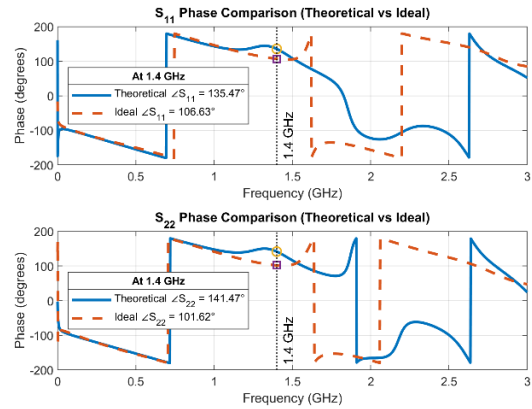


Figure 8: Comparison of S_{11} & S_{22} Phase

Figure 7 shows that at 1.4 GHz, both S_{11} and S_{22} are about -34 to -39 dB, an excellent impedance matching condition at both ports with the minimum reflections seen in were confirmed. The good concurrence of the simulated and optimal curves around the target frequency demonstrates that the microstrip line exhibits almost ideal characteristics at 1.4 GHz, which can guarantee high-power transmission efficiency.

Figure 8 shows that at 1.4 GHz, the phases of S_{11} and S_{22} for theoretical and ideal curves are very close, indicating consistent phase behavior at both ports, confirming that reflections are not only small in magnitude but also stable in phase at the design frequency, supporting proper two-port matching.

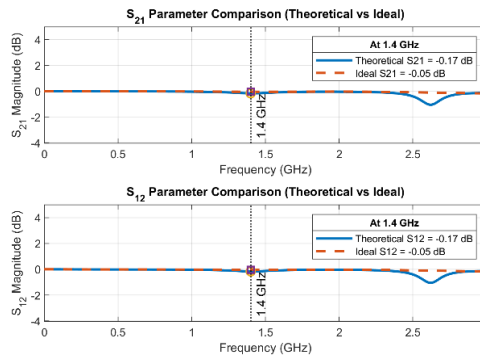


Figure 9: Comparison of S_{21} & S_{12} Magnitude

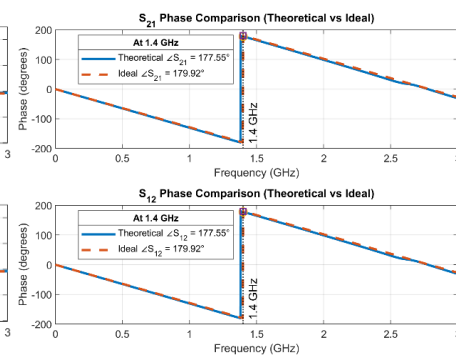


Figure 10: Comparison of S_{21} & S_{12} Phase

The Figure 9 shows the magnitudes of S_{21} and S_{12} remain close to 0 dB at 1.4 GHz, indicating low insertion loss and efficient power transmission through the microstrip line. The ideal design matches the requirement to have the maximum value for S_{21} and validates our design.

Figure 10 shows the phases of S_{21} and S_{12} are nearly identical for both theoretical and ideal cases, indicating reciprocal and stable phase transmission. This confirms that our design for microstrip line is valid because the electrical length was set to be 180° .

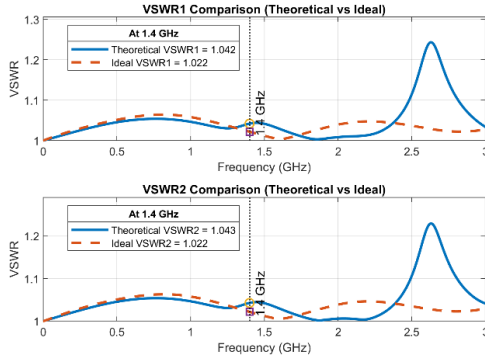


Figure 11: Comparison of VSWR1 and VSWR2

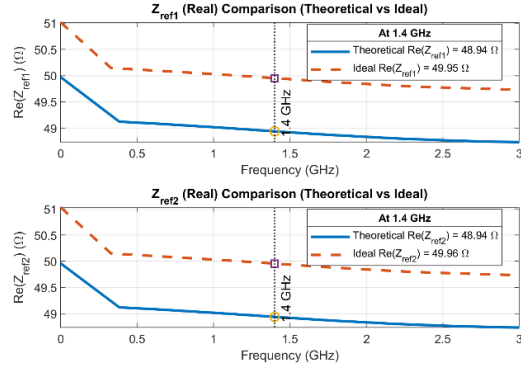


Figure 12: Comparison of Zref(1) and Zref(2)

Figure 11 presents that both VSWR1 and VSWR2 are very close to 1 at 1.4 GHz (≈ 1.02 – 1.04), confirming a great impedance matching at both ports. The low VSWR values are consistent with the very low S_{11} and S_{22} magnitudes, indicating minimal standing waves and efficient power transfer at the target frequency.

Figure 12 shows that at 1.4 GHz, the real parts of both Z_{ref1} and Z_{ref2} are close to 50Ω (≈ 48.9 – 49.9Ω) and shows that good impedance matching to the reference system impedance. The close agreement between theoretical and ideal values confirms that both ports are properly matched at the design frequency.

2. Loaded Microstrip (Without Stub)

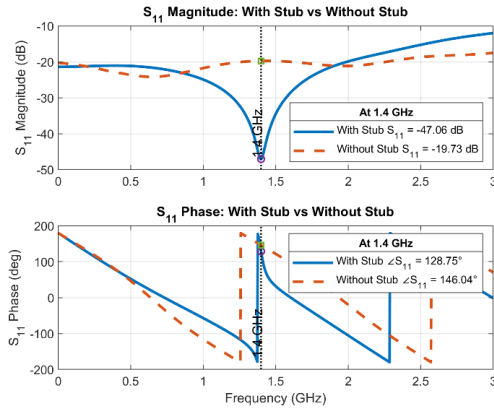


Figure 13: Comparison of S_{11} Magnitude and Phase for with & without stub

Figure 13 shows that, at 1.4 GHz, there is a mismatch between the line and the load before adding the shunt stub. After adding a single shunt stub, the value drops to -47 dB, meaning the stub strongly improves the matching by reducing reflection at the target frequency. This is also visible in the phase plot. A sharp change in phase is observed around 1.4 GHz, as the value approaches zero in the region where matching is very good with the stub.

1; this corresponds to an almost ideal match. Therefore, single-stub shunt tuning cancels the reactive part created by the load at the design frequency, brings the input impedance closer to 50Ω , and minimizes the mismatch.

In Figure 14, the VSWR plot clearly shows how much the shunt stub improves the matching at the target frequency. When the stub is added, the VSWR drops to

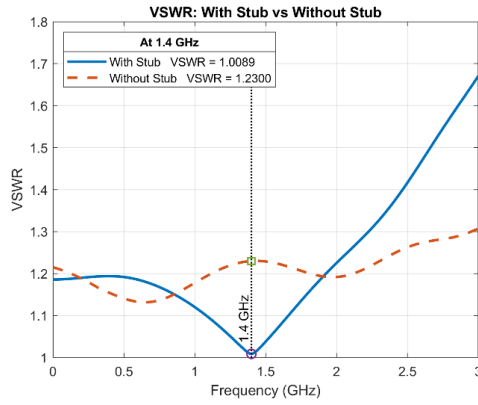


Figure 14: Comparison of VSWR for with & without stub

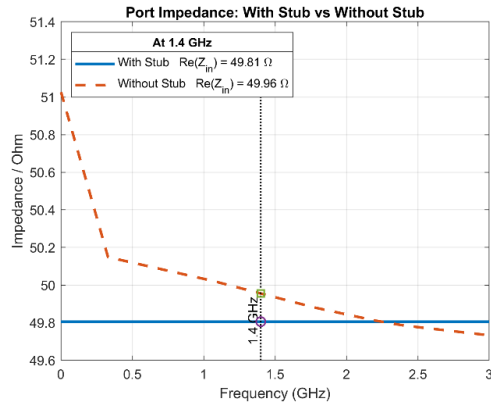


Figure 15: Comparison of Port Impedance for with & without stub

In Figure 15, this plot shows the real part of the port impedance, and it can be seen that at 1.4 GHz the value is very close to 50 Ω in both cases. This result shows that adding the stub does not create a major change in the system.

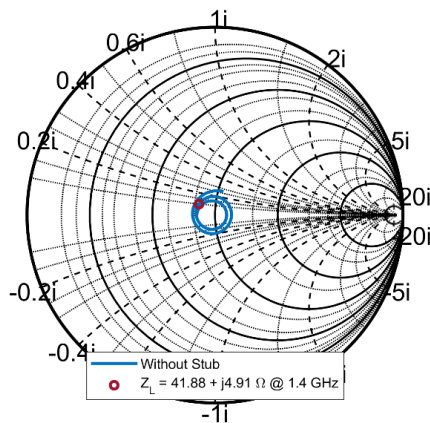


Figure 13: Smith Chart of the without stub loaded

In Figure 16, the Smith Chart shows the mismatch at the input before adding the stub. At 1.4 GHz, the marked point shows that the real part of the impedance stays clearly below 50 Ω . It was observed that it reached 42.11 Ω . This deviation shows that the mismatch cannot be removed without adding the stub.

As a result, this clearly shows the effect of single shunt stub matching on the microstrip line at 1.4 GHz. The complex load connected at the end causes a clear mismatch. At 1.4 GHz, the input impedance deviates from the ideal value and leads to a certain level of reflection.

3. Single-Stub Shunt Tuning Theoretical and Optimization

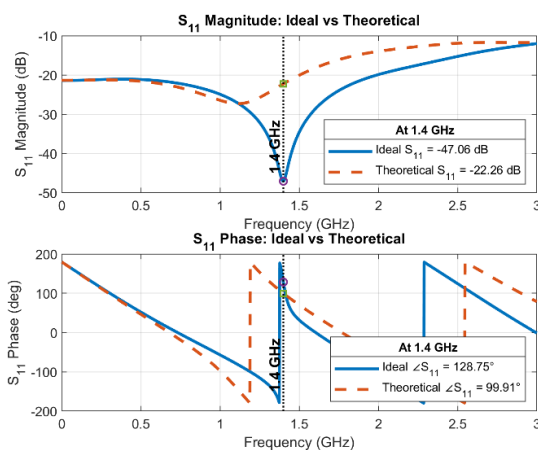


Figure 14: Comparison of S11 Magnitude and Phase for with stub (Ideal & Theoretical)

In Figure 17, the theoretically calculated single-stub shunt tuning was optimized to achieve impedance matching at 1.4 GHz. The theoretical design shows that the minimum reflection point did not fit exactly at 1.4 GHz, so the matching was limited. After the shunt stub optimization, S_{11} decreased to -47.06 dB. This confirms that the reflection coefficient was reduced and the matching became almost ideal. The phase curve also shows that, by shifting the stub, the reactive component at the input was balanced more effectively.

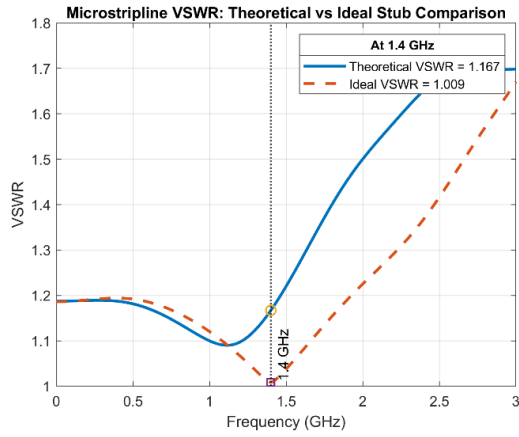


Figure 18: Comparison of VSWR for with stub (Ideal & Theoretical)

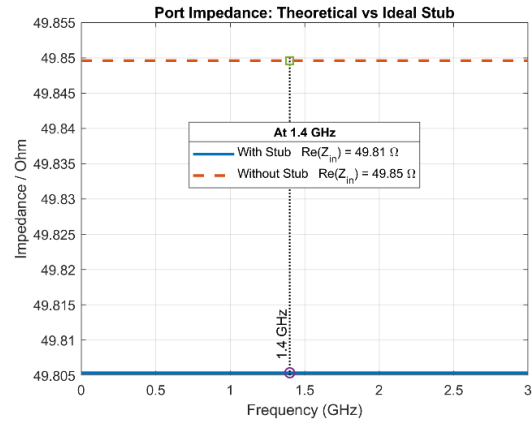


Figure 19: Comparison of Port Impedance for with stub (Ideal & Theoretical)

Figure 18 clearly shows the effect of the optimization in the VSWR comparison. Although the theoretical stub result is already good, after optimization the VSWR drops to 1.009 and an almost ideal match is achieved. This result confirms that the optimization moves the minimum reflection point to the target frequency and cancels the reactive component caused by the load much more effectively at this frequency.

In Figure 19, at 1.4 GHz the difference between the theoretical and the optimized case is only 0.04Ω , which is very small. This result shows that the main effect of the optimization is to cancel the imaginary part of the input impedance better at 1.4 GHz and bring it closer to 50Ω .

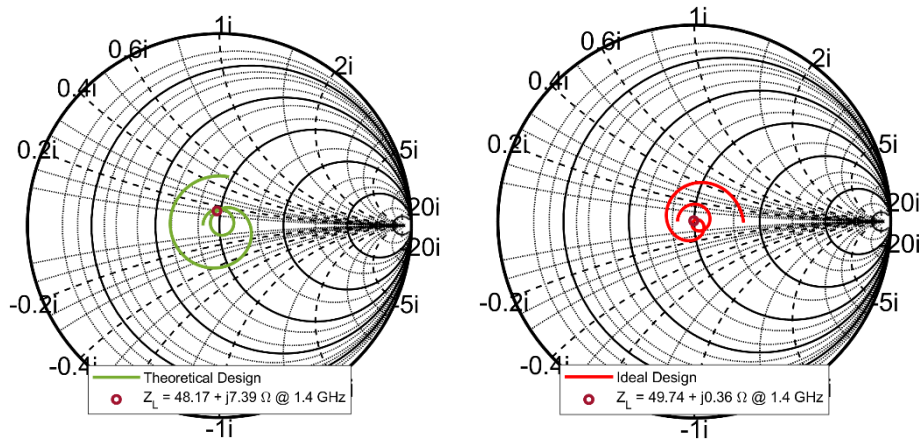


Figure 20: Smith Chart of The Theoretical & Ideal Stub at 1.4 GHz

In Figure 20, the Smith Chart shows how the input impedance of the theoretical and optimized stub changes with frequency. The theoretical impedance value is 48.55Ω , but after optimization it was observed to be 49.53Ω . In the theoretical design, at 1.4 GHz the input impedance stays above the center point on the Smith Chart. After optimization by shifting the stub position on the microstrip line, the point on the Smith chart moves almost exactly to the center. In this way, it is seen that the matching reaches an almost ideal level.

Overall, by adding the shunt stub, the matching at the design frequency is improved significantly by compensating the reactive part and moving the operating point toward the center of the chart. After optimizing the circuit, the impedance becomes almost 50Ω at 1.4 GHz. It also minimizes the reflection.

CONCLUSIONS

In this project, a $50\ \Omega$ microstrip line on an FR-4 substrate was designed for 1.4 GHz and evaluated under a complex load. The unloaded line design provided a characteristic impedance very close to the target value, which confirms the accuracy of the analytical sizing and the CST modelling. When the complex load was connected without a matching network, a clear deviation in the input impedance was observed. To remove this mismatch, an open-circuit shunt stub was used, because it can provide the susceptance needed to cancel the reactive part at the design frequency as a simple open-ended microstrip section. After optimizing the stub position, the matching performance improved clearly. Overall, the results prove that the open-circuit single shunt stub network is an effective and practical method for microstrip matching.

REFERENCES

- [1] Pozar, D. M. (2011). *Microwave engineering: Chapter 3&5*. John Wiley & Sons.